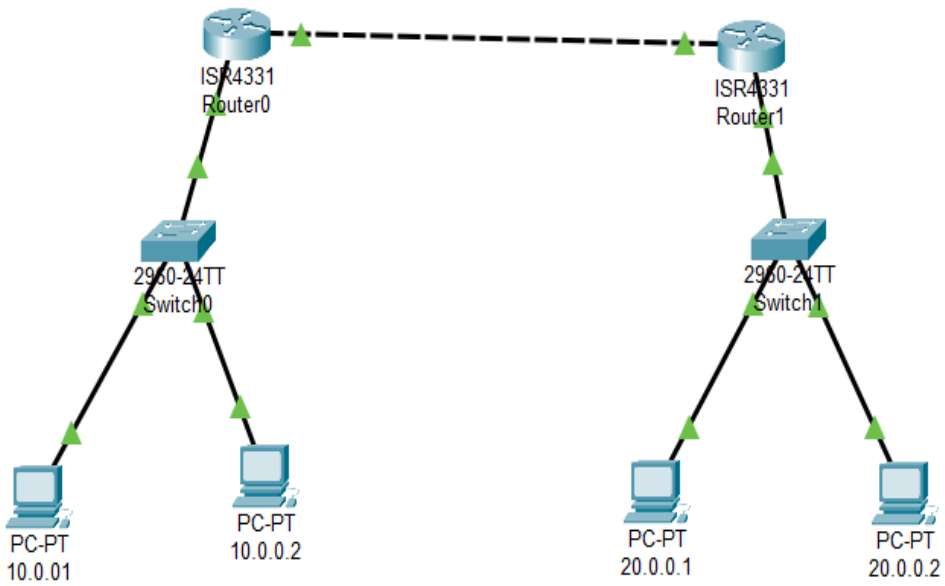
Date: 27/08/'26

Lab Practical #12:

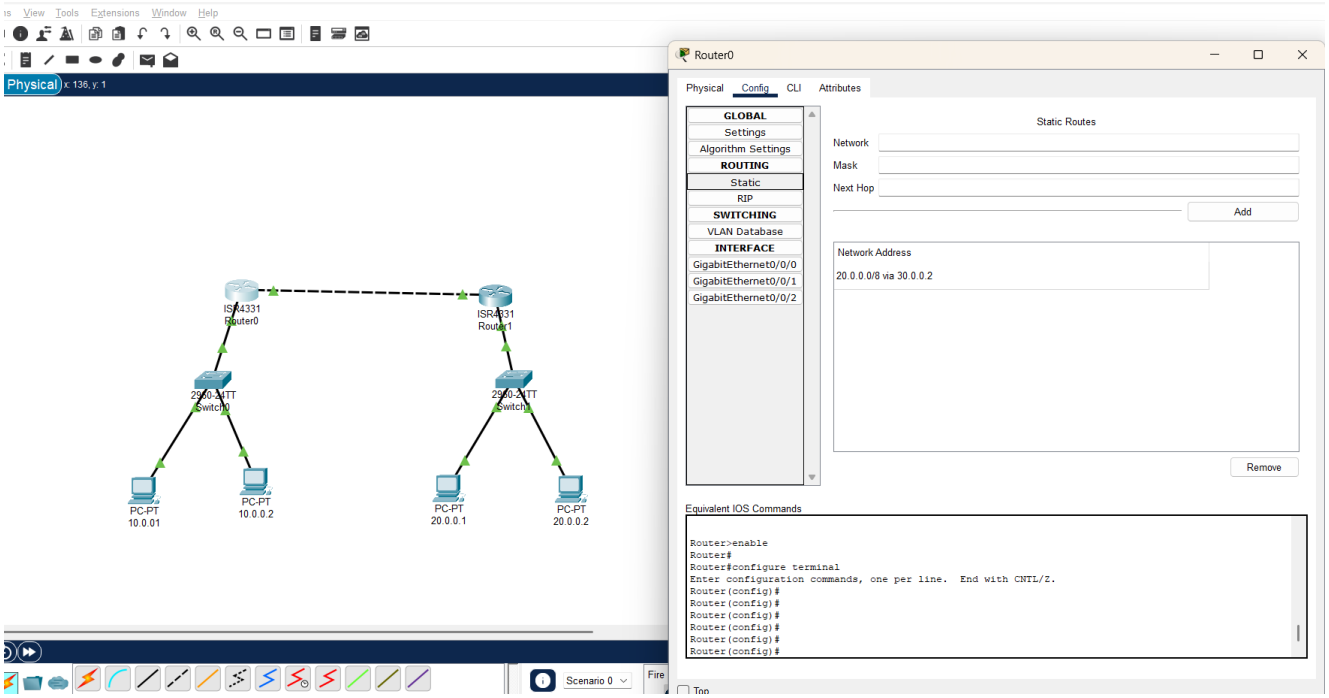
Study the concept of routing using packet tracer. (Static Routing)

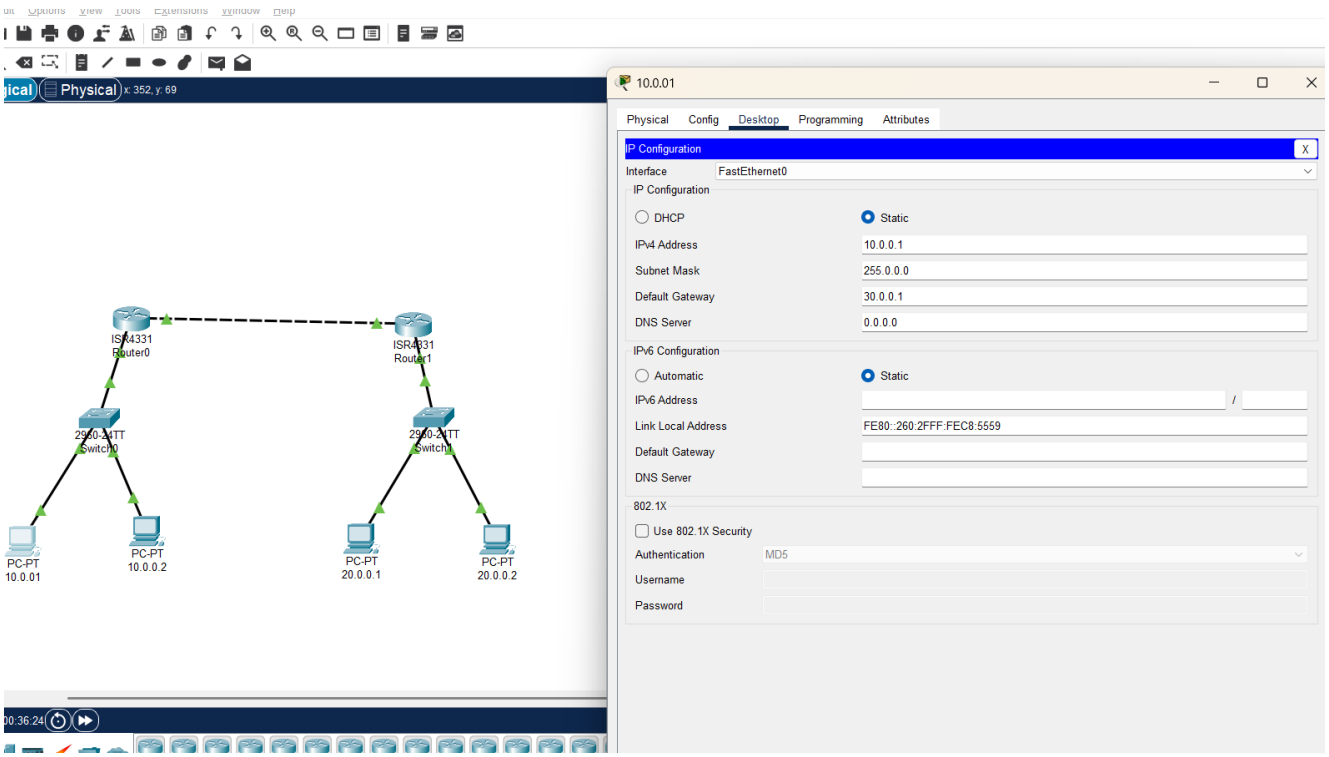
Practical Assignment #12:

A. Connect the two different networks based on the calculated IP addresses and subnet using a packet tracer.

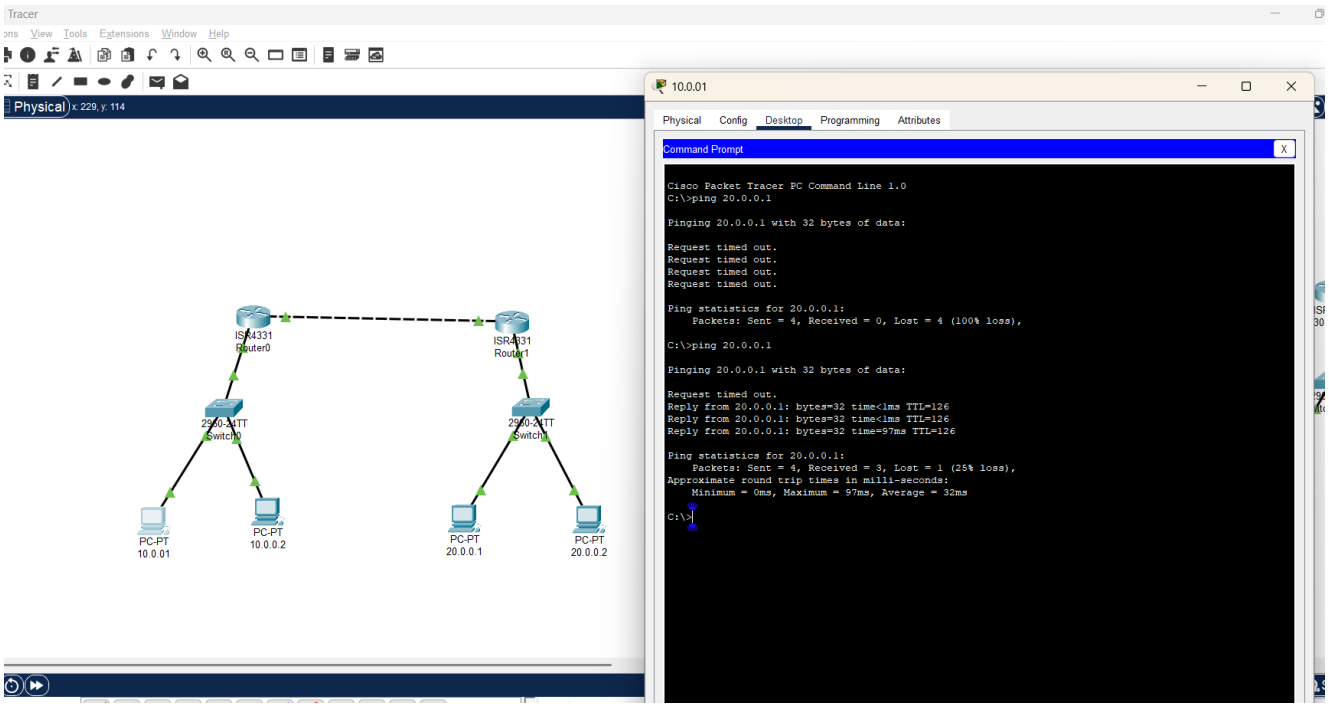


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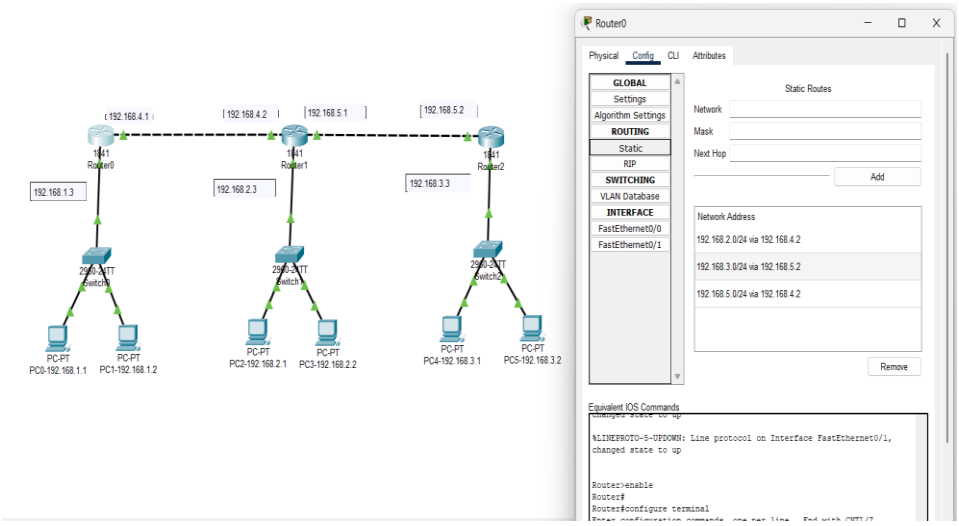

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0/0
GigabitEthernet0/0/1
GigabitEthernet0/0/2
Static Routes
Network
Mask
Next Hop
Add
Network Address
20.0.0.0/8 via 30.0.0.2
Remove
Equivalent IOS Commands
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
ISR4331 Router0
ISR4331 Router1
2950-24TT Switch0
2950-24TT Switch1
PC-PT 10.0.0.1
PC-PT 10.0.0.2
PC-PT 20.0.0.1
PC-PT 20.0.0.2


10.0.01
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
IPv4 Address
10.0.0.1
Subnet Mask
255.0.0.0
Default Gateway
30.0.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::260:2FFF:FEC8:5559
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
ISR4331 Router0
ISR4331 Router1
2950-24TT Switch0
2950-24TT Switch1
PC-PT 10.0.0.1
PC-PT 10.0.0.2
PC-PT 20.0.0.1
PC-PT 20.0.0.2

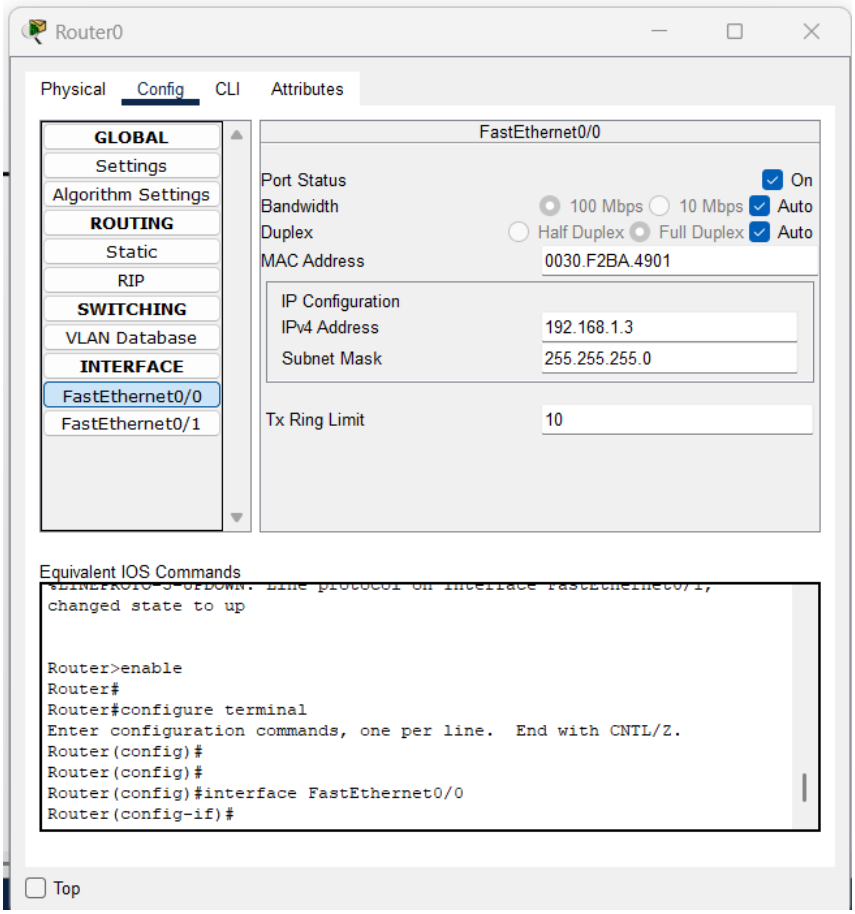
Date: 27/08/'26

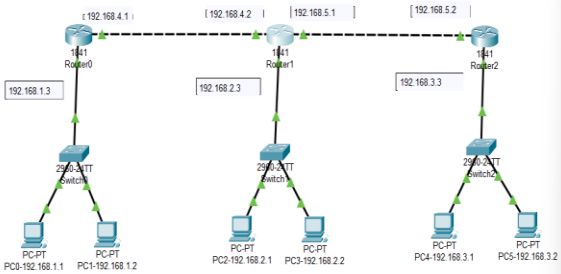


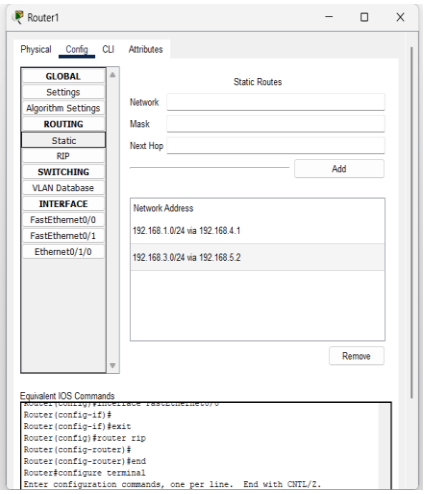
B . Connect the three different networks based on the calculated IP addresses and subnet using a packet tracer.



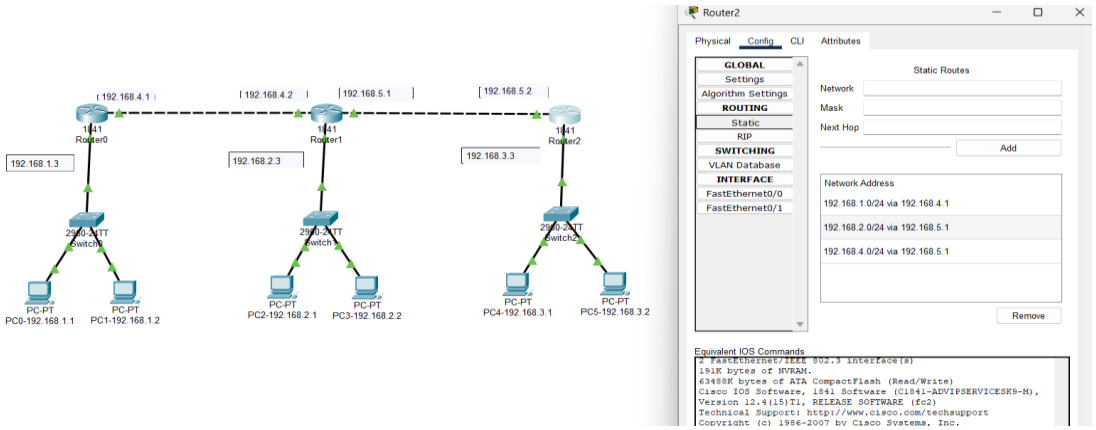
Date: 27/08/'26

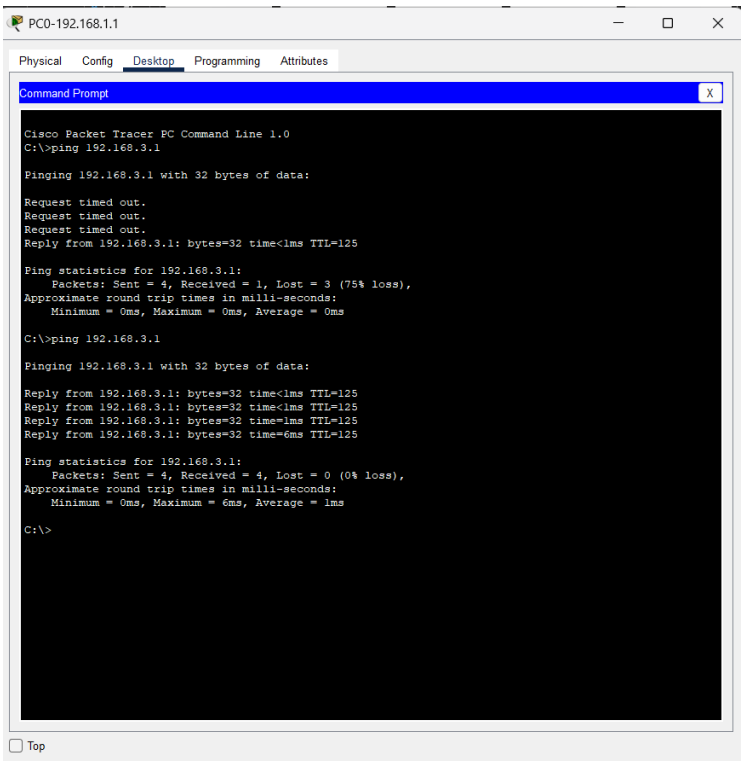

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
FastEthernet0/0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0030.F2BA.4901
IP Configuration
IPv4 Address
192.168.1.3
Subnet Mask
255.255.255.0
Tx Ring Limit
10
Equivalent IOS Commands
changed state to up
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
Top


192.168.4.1
192.168.4.2
192.168.5.1
192.168.5.2
Router0
Router1
Router2
192.168.1.3
192.168.2.3
192.168.3.3
PC0-192.168.1.1
PC1-192.168.1.2
PC2-192.168.2.1
PC3-192.168.2.2
PC4-192.168.3.1
PC5-192.168.3.2


Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Ethernet0/1/0
Static Routes
Network
Mask
Next Hop
Add
Network Address
192.168.1.0/24 via 192.168.4.1
192.168.3.0/24 via 192.168.5.2
Remove
Equivalent IOS Commands
Router(config-if)#
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#
Router(config-router)#end
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

Date: 27/08/'26


192.168.4.1
192.168.4.2
192.168.5.1
192.168.5.2
Router0
Router1
Router2
192.168.1.3
192.168.2.3
192.168.3.3
Switch0
Switch1
Switch2
PC0-192.168.1.1
PC1-192.168.1.2
PC2-192.168.2.1
PC3-192.168.2.2
PC4-192.168.3.1
PC5-192.168.3.2
Router2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Static Routes
Network
Mask
Next Hop
Add
Network Address
192.168.1.0/24 via 192.168.4.1
192.168.2.0/24 via 192.168.5.1
192.168.4.0/24 via 192.168.5.1
Remove
Equivalent IOS Commands
191K bytes of NVRAM.
63488K bytes of ATA CompactFlash (Read/Write)
Cisco IOS Software, 1841 Software (C1841-ADVIPSERVICESK9-M),
Version 12.4(15)T1, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.


PC0-192.168.1.1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.1
Pinging 192.168.3.1 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Reply from 192.168.3.1: bytes=32 time<1ms TTL=125
Ping statistics for 192.168.3.1:
Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.3.1
Pinging 192.168.3.1 with 32 bytes of data:
Reply from 192.168.3.1: bytes=32 time<1ms TTL=125
Reply from 192.168.3.1: bytes=32 time<1ms TTL=125
Reply from 192.168.3.1: bytes=32 time=1ms TTL=125
Reply from 192.168.3.1: bytes=32 time=6ms TTL=125
Ping statistics for 192.168.3.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 6ms, Average = 1ms
C:\>
Top